Foundation Parameters

Health Hub Business Plan v2 | Shared Assumptions Document Version: 2.0 | **Date:** March 2026

> *Every number, timeline, and cost figure in the v2 business plan traces back to this document. When a downstream document says "per params.md," it means here.*

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1. Company Overview

Health Hub is an international health technology company building a multi-tenant, multi-stakeholder healthcare platform. The platform serves patients, GPs, specialists, pharmacies, diagnostic labs, and administrators — each with dedicated workflows, dashboards, and service layers.

Current state: Functional prototype (Angular 21 SSR + Express 5 + PostgreSQL 16

- WebSocket + Daily.co video). The prototype can serve a limited number of users with manual support. It is not yet hardened for security or scale.

Primary launch geography: East Africa (Ethiopia first, Kenya second).

Business model: Transaction-first health marketplace with admin-assisted care navigation. The company operates as both a technology platform and an assisted healthcare delivery operation.

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2. Team Composition and Capacity

2.1 Core Team (3 Members)

Role	Description	Hours/Day	Hours/Month	Status
Owner / Founder	Idea owner, product vision, strategic decisions, investor relations	4	~88	Part-time
Founding Member 1	Operations and management, partner relations, business development	4	~88	Part-time
Founding Member 2	Technical head, product testing, architecture oversight, QA leadership	4	~88	Part-time

Core team monthly capacity: ~264 hours (3 people x 88 hours)

2.2 Development Team (7 Part-Time Developers)

Sub-team	Members	Roles	Hours/Day Each	Combined Hours/Month
Android / Backend team	3-4	Building the patient-facing Android APK and supporting backend work	4	~308-352
Web Platform team	2	1 DevOps engineer + 1 Full-stack developer for web hardening	4	~176
Shared / Flex	1	Crosses between Android and web as needed	4	~88

Developer team monthly capacity: ~572 hours (7 people x ~82 productive hours after coordination overhead)

2.3 Advisory Board (2 Members)

Role	Contribution	Hours/Month
Advisor 1	Finance, compliance, regulatory strategy	~10-15
Advisor 2	Strategic review, market guidance, investor introductions	~10-15

Advisory monthly capacity: ~20-30 hours

2.4 Capacity Summary

Level	Total Productive Technical Hours/Month	Explanation
Level 1 (Bootstrapped)	~320 hrs/mo	Lean utilization; more sequencing, founder time split across responsibilities
Level 2 (Ideal)	~480 hrs/mo	Balanced baseline; enough concurrency for build + operating prep

Level 3 (Fully Funded)	~640 hrs/mo	Full effective utilization of all part-time contributors

Key rule: For paired roles (e.g., 2 people sharing one function), capacity is modeled as equivalent to a single full-time person. This reflects part-time scheduling, handoff overhead, and coordination cost.

Important: No team member is a full-time employee. All contributors work part-time (~4 hours/day, ~22 working days/month = ~88 hours/month per person).

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3. Rate Card

3.1 Product and Delivery Rates

Role Category	USD/hr	INR/hr	Use Case
Junior developer	$10	~INR 830	Routine implementation, test support, basic frontend/backend tasks
Senior developer	$20	~INR 1,660	Full-stack ownership, DevOps, architecture, hardening, complex features
Management / Technical head	$20	~INR 1,660	Founder office, product management, testing oversight, strategic work
Everyone else (blended)	$15	~INR 1,245	Android team (blended), design, general contributors, advisory

3.2 Blended Rate Calculation

Team Segment	Rate	Members	Monthly Cost
Core team (3 x management)	$20/hr	3	3 x 88 x $20 = $5,280
Senior developers (DevOps + Full-stack)	$20/hr	2	2 x 88 x $20 = $3,520
Junior developers (Android/backend)	$10/hr	3	3 x 88 x $10 = $2,640
Blended contributors (flex + Android lead)	$15/hr	2	2 x 88 x $15 = $2,640
Advisory board	$15/hr	2	2 x 25 x $15 = $750

Total monthly team economic cost: ~$14,830/month

Blended average across all contributors: ~$14.50/hr

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4. Operational Staffing and Employed Coverage

4.1 Salary Benchmarks (India-Based Ops Staff)

These are employees hired to run the live healthcare operation. They are separate from the product/development team.

Role	INR/Month	USD/Month	Basis
MBBS Doctor (telemedicine)	60,000	~$720	Telemedicine doctor benchmarks in India: INR 40,000-80,000/mo (Practo, NHM anchors); INR 60,000 is mid-range for part-time/shift-based tele-consult doctors
Admin Ops / Callback Staff	18,000	~$217	Indian customer support and back-office benchmarks for semi-skilled roles
Technical Support (L1/L2)	25,000	~$301	Junior technical support / helpdesk benchmarks in Indian metros

Source: Web research confirms MBBS telemedicine doctor salaries range from INR 40,000-80,000/month in India (2025-2026 data from Practo benchmarks, NHM Medical Officer bands, and health tech platform ranges). The INR 60,000 midpoint aligns with part-time / shift-based remote consultation work.

4.2 Minimum Employed Coverage Ladder

Phase	Doctors	Admin Ops	Technical Support	Monthly INR	Monthly USD
Pre-Pilot	0	0	0	0	$0
Pilot 1 (up to 1,000 users)	2	2	2	206,000	~$2,482
Pilot 2 (5,000-10,000 users)	4	4	2	362,000	~$4,361
Production Readiness	6	6	3	543,000	~$6,542
Production (full scale)	8-10	8	4	748,000-868,000	~$9,012-$10,458

4.3 Why These Staff Are Required

Role	Why Required	What They Do
MBBS Doctors	Platform must guarantee minimum clinical coverage for continuity and service confidence, independent of partner-clinic doctors	Handle scheduled and on-demand tele-consultations, triage callbacks, clinical oversight
Admin Ops	The operating model is admin-assisted; callbacks, exception handling, and partner coordination cannot be automated in early phases	Callback handling, tourism/travel follow-up, manual prescription/lab fulfilment, partner coordination, patient onboarding assistance
Technical Support	Live users create real issue-response obligations; the dev team cannot absorb support load and still build	L1 issue triage, basic troubleshooting, escalation to dev team, live-ops monitoring, user guidance

4.4 Coverage Math

For near-24/7 coverage (which production requires for a healthcare platform):

Shift Pattern	Doctors Needed	Admin Needed	Tech Support Needed
8-hour shift, 7 days/week, 2 shifts/day	4 minimum (2 per shift)	4 minimum	2 minimum
3-shift 24/7 coverage	6 minimum	6 minimum	3 minimum
24/7 + buffer for leave/training	8-10	8	4

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5. Android APK Scope

Parameter	Value
Total hours	480 hours
Team	3-4 developers from the Android/backend sub-team
Average blended rate	$15/hr
Total cost	$7,200
Status	Fixed external quote; already contracted
Scope	Patient-facing mobile application covering registration, dashboard, consultation request, video/audio/chat, prescription viewing, lab results, notifications, payments
Target	Core patient journeys ready by Pilot 1; full scope by Pilot 2

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6. Web Platform Scope

6.1 Effort Estimation Basis

The remaining web platform work is estimated at **150-160% of Android effort**, based on a real estimate from the development team.

Parameter	Value	Derivation
Android effort	480 hours	Fixed quote
Web multiplier	1.55x (midpoint of 1.5x-1.6x)	Real team estimate; web has more role surfaces but benefits from existing prototype
Remaining web effort	**744 hours** (range: 720-768)	480 x 1.55 = 744
Modeled expected case	**745 hours**	Rounded for planning

6.2 What This Covers

The 745-hour web estimate includes:

- Security hardening and production-readiness fixes
- Role-flow completion (GP, specialist, pharmacy, diagnostics, admin)
- SSR safety, auth hardening, rate limiting
- Analytics, reporting dashboards, partner tooling
- API hardening, error handling, monitoring integration
- Performance optimization and load testing
- Release governance tooling

6.3 What This Does Not Cover

- Android APK (separate 480-hour workstream)
- Doctor/admin training content creation (see Section 17)
- SOP writing and operational documentation (see Section 17)
- Partner onboarding materials (see Section 17)

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7. Infrastructure and Operational Costs

7.1 Technology Infrastructure

Phase	Hosting & Monitoring	Video (Daily.co)	Database	Total Tech Infra
Pre-Pilot	$50-80/mo	$0-20/mo	$15-30/mo	$65-130/mo
Pilot 1	$80-150/mo	$50-100/mo	$30-50/mo	$160-300/mo
Pilot 2	$150-300/mo	$100-200/mo	$50-100/mo	$300-600/mo
Production	$300-500/mo	$200-400/mo	$100-200/mo	$600-1,100/mo

7.2 Operational Infrastructure (NEW in v2)

Item	Phase Introduced	Monthly Cost	Notes

Admin phones (2-4 devices)	Pre-Pilot	$0 (one-time $200-400)	Android phones for admin staff to handle callbacks
Phone top-ups / SIM costs	Pilot 1	$30-60/mo	Call minutes and data for admin callback operations
Telecom / VoIP for callbacks	Pilot 1	$50-100/mo	Cloud telephony or VoIP for systematic callback management
Call center software (basic)	Pilot 2	$50-100/mo	Basic CRM/ticketing for callback tracking
Additional devices for scaling	Pilot 2	$0 (one-time $300-600)	More phones/tablets as admin team grows
Support ticketing tools	Pilot 1	$0-30/mo	Free tier initially, paid as volume grows

7.3 One-Time Equipment Costs

Item	Cost	When
Admin phones (initial batch of 2)	$200-300	Pre-Pilot
Admin phones (expansion to 4-6)	$300-500	Pilot 2
Laptops/workstations for support staff	$400-600	Pilot 1
Miscellaneous office/ops setup	$200-300	Pilot 1

7.4 Combined Infrastructure Ranges (Monthly)

Phase	Tech Infra	Ops Infra (recurring)	Combined Monthly
Pre-Pilot	$65-130	$0-20	$65-150
Pilot 1	$160-300	$110-220	$270-520
Pilot 2	$300-600	$160-320	$460-920
Production	$600-1,100	$200-400	$800-1,500

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8. Equity and Capital Structure

8.1 Cap Table — Three Pools

Pool	Share	Description
Owner / Founder	60%	Sole idea owner and primary decision maker. Controls voting majority.
Founding Team + Strategic Partners	20%	2 founding members + any future strategic partner (e.g., key clinic chain, technology partner). Vested over 3-4 years with 1-year cliff.
Investor Pool	20%	Reserved ceiling for outside capital in this planning phase. Drawn from as investor rounds close.

8.2 Dilution Logic

- The 20% investor pool is the **ceiling** for outside capital during the current planning phase (through Production launch).
- If the investor requires more than 20%, the dilution comes proportionally from the other two pools (founder and founding team).
- Anti-dilution: founder maintains majority control (>50%) through Production in all modeled scenarios.
- The founding team pool includes both current founding members and a small reserve (~3-5%) for future strategic hires or partnerships.

8.3 Equity Implications by Scenario

Scenario	Founder Post-Raise	Founding Team Post-Raise	Investor Post-Raise
S1 (no investor)	60%	20%	0% (pool reserved)
S2 (investor during transition)	55-60%	18-20%	15-20%
S3 (investor after Pilot 1)	52-58%	17-19%	18-20%
If investor demands >20%	Negotiated	Negotiated	20-30% max

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9. East Africa Market Parameters

9.1 Ethiopia

Parameter	Value	Source
Population	~130 million	World Bank 2025
GDP per capita	~$1,020	World Bank 2025
Health expenditure per capita	~$28/year	WHO 2024
Smartphone penetration (urban)	35-45%	GSMA 2025
Addis Ababa population	~5.5 million	Census estimate
Private clinics in Addis	2,000-3,000	Ministry of Health estimates

Mobile money (Telebirr) users	~40 million	Ethio Telecom reports
Regulatory environment	Telecom liberalization ongoing; health tech regulation nascent	MOH digital health strategy 2025

9.2 Kenya

Parameter	Value	Source
Population	~56 million	World Bank 2025
GDP per capita	~$2,100	World Bank 2025
Health expenditure per capita	~$84/year	WHO 2024
Smartphone penetration (urban)	55-65%	GSMA 2025
Nairobi population	~5.0 million	Census estimate
M-Pesa users	~33 million	Safaricom reports
Regulatory environment	More mature digital health regulation; PPB oversight	Kenya Health Act

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10. Currency and FX Assumptions

Currency Pair	Rate Used	Buffer Applied	Notes
USD to INR	83.0	—	Stable corridor; used for Indian ops staff costing
USD to ETB	57.0	10-15% downward	Ethiopian birr depreciating; buffer protects projections
USD to KES	130.0	10% downward	More stable than ETB but still volatile

All revenue figures in downstream documents are converted to USD with the FX buffer applied to account for depreciation risk.

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11. Revenue Model Assumptions

11.1 Revenue Streams

Stream	Type	When Active
GP consultation fees	Per-transaction (B2C)	Pilot 1 (subsidized) onward
Specialist consultation fees	Per-transaction (B2C)	Pilot 2 onward
Pharmacy commission	Take-rate on order value (B2B)	Pilot 1 onward
Diagnostics commission	Take-rate on order value (B2B)	Pilot 1 onward
Care coordination / travel	Per-case fee	Pilot 2 onward
Facility platform fees	Monthly SaaS (B2B)	Production onward
Premium patient subscription	Monthly (B2C)	Production onward

11.2 Pricing Corridors

Service	Ethiopia Benchmark	Kenya Benchmark
GP consult	ETB 150-300 ($2.60-5.25)	KES 300-500 ($2.30-3.85)
Specialist consult	ETB 300-600 ($5.25-10.50)	KES 500-1,000 ($3.85-7.70)
Pharmacy take-rate	8-12%	8-12%
Diagnostics take-rate	10-15%	10-15%
Facility platform fee	ETB 2,000-5,000/mo	KES 5,000-10,000/mo
Patient premium sub	ETB 200-400/mo	KES 400-800/mo

11.3 Revenue Timing

Phase	Revenue Expectation
Pre-Pilot	$0 — internal testing only
Pilot 1	$0-500/mo — subsidized/symbolic pricing, learning phase
Pilot 2	$500-3,000/mo — 50% of target pricing, real monetization experiments
Production	$3,000-15,000/mo — full pricing, scaling user base

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12. Success Metric

Primary: Cumulative total revenue exceeds cumulative total capital investment.

Secondary metrics:

- Monthly active patients > 1,000 by end of Pilot 2
- LTV:CAC ratio > 3x by Production
- Monthly burn rate sustainable for 6+ months at any point
- Doctor utilization rate > 60% during operational hours

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13. Phase Definitions

Phase	User Scale	Duration Range	Primary Focus
Pre-Pilot	100-200 (internal)	3-6 months	Internal testing with mock and real team members. Android APK under construction. Web hardening. Doctor/admin training design. SOPs drafted.
Pilot 1	Up to 1,000	2-3 months	First external pilot in Ethiopia. Live callbacks, real patients, partner clinics. Android APK core ready.
Pilot 2	5,000-10,000	3-13 months	Larger pilot via marketing. Application hardening in background. Kenya prep. Staffing ladder active.
Production Readiness	Full scale	4-12 months	Final hardening, compliance, launch governance, support schedules, production release.

Total timeline range: 16-34 months depending on scenario and investment level.

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14. Scenario and Investment Level Definitions

14.1 Three Scenarios

Scenario	Description	Self-Funded Runway
S1 — Fully Self-Funded	No external investor through Production. All costs from founders.	Month 0 through Production
S2 — Investor During Transition	Self-fund through Pilot 1 into Pilot 2 transition. Investor arrives during Pilot 1→2 transition or after Pilot 2.	3-12 months
S3 — Investor After Pilot 1	Self-fund only through Pre-Pilot and into Pilot 1. Investor arrives right after Pilot 1 completes.	0-3 months

14.2 Three Investment Levels

Level	Label	Budget vs Ideal	Description
L1	Bootstrapped / Ad Hoc	~40% of ideal	Maximum constraint. Bare minimum spend. Slower everything.
L2	Ideal / Basic	100% (baseline)	Minimum viable investment. Just enough to meet milestones. Moderate breathing room.
L3	Fully Funded / Corporate	150-200%+ of ideal	Full corporate approach. Proper resourcing at every stage.

14.3 Mix-and-Match Rule

For Scenarios 2 and 3, our own spend level and the investor's contribution level can differ. This creates:

Scenario	Combinations	Notation
S1	3 (our level only)	S1-L1, S1-L2, S1-L3
S2	9 (our level x investor level)	S2-O1-I1 through S2-O3-I3
S3	9 (our level x investor level)	S3-O1-I1 through S3-O3-I3
Total	**21 combinations**	

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15. Full Combination Matrix

#	Code	Our Level	Investor Level	Investor Timing
1	S1-L1	Bootstrapped	None	Never
2	S1-L2	Ideal	None	Never
3	S1-L3	Fully Funded	None	Never
4	S2-O1-I1	Bootstrapped	Bootstrapped	Pilot 1→2 transition
5	S2-O1-I2	Bootstrapped	Ideal	Pilot 1→2 transition
6	S2-O1-I3	Bootstrapped	Fully Funded	Pilot 1→2 transition
7	S2-O2-I1	Ideal	Bootstrapped	Pilot 1→2 transition
8	S2-O2-I2	Ideal	Ideal	Pilot 1→2 transition
9	S2-O2-I3	Ideal	Fully Funded	Pilot 1→2 transition
10	S2-O3-I1	Fully Funded	Bootstrapped	Pilot 1→2 transition
11	S2-O3-I2	Fully Funded	Ideal	Pilot 1→2 transition

12	S2-O3-I3	Fully Funded	Fully Funded	Pilot 1→2 transition
13	S3-O1-I1	Bootstrapped	Bootstrapped	After Pilot 1
14	S3-O1-I2	Bootstrapped	Ideal	After Pilot 1
15	S3-O1-I3	Bootstrapped	Fully Funded	After Pilot 1
16	S3-O2-I1	Ideal	Bootstrapped	After Pilot 1
17	S3-O2-I2	Ideal	Ideal	After Pilot 1
18	S3-O2-I3	Ideal	Fully Funded	After Pilot 1
19	S3-O3-I1	Fully Funded	Bootstrapped	After Pilot 1
20	S3-O3-I2	Fully Funded	Ideal	After Pilot 1
21	S3-O3-I3	Fully Funded	Fully Funded	After Pilot 1

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16. Existing Asset Inventory

16.1 Codebase (Verified from Repository)

Asset	Detail	Replacement Value
Angular 21 SSR web application	10 feature modules (auth, dashboard, patient, GP, specialist, pharmacy, diagnostics, admin, notifications, shared)	$18,000-24,000
Express 5 API server	16 API route files covering all roles + health endpoints	$6,000-8,000
PostgreSQL schema	12 migration files, comprehensive multi-role schema	$3,000-4,000
WebSocket layer	Real-time notifications, queue updates, chat infrastructure	$2,000-3,000
Daily.co video integration	Video consultation infrastructure (placeholder/scaffold)	$1,500-2,000
Auth and security layer	JWT, bcrypt, role guards, interceptors, SSR safety	$2,000-3,000
DevOps and deployment	Render config, health endpoints, rate limiting, structured logging	$1,000-1,500

Total estimated replacement cost: $33,500-$45,500

16.2 Documentation and Planning Assets

- CLAUDE.md audit system with 19 tracked issues (all resolved)
- 11-agent orchestration system for project management
- Business plan v1 (15 documents + scenarios + competitive analysis)
- Agent boundary configurations and task boards

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17. Parallel Operational Workstreams

These workstreams run alongside software development and are a critical part of building the company — not just the product. They explain why reduced coding hours do not automatically collapse the timeline.

17.1 Workstream Inventory

Workstream	Starts	Continues Through	Effort (Hours)	Owner
Doctor training protocols and playbooks	Pre-Pilot	Production	120-160	Founding Member 1 + Medical Advisor
Admin callback SOPs and escalation trees	Pre-Pilot	Production	80-120	Founding Member 1
Admin operational guidelines (daily workflows)	Pre-Pilot	Pilot 2	60-80	Founding Member 1
Technical support runbooks	Pilot 1	Production	60-80	Technical Head
Partner onboarding kits (clinics, labs, pharmacies)	Pre-Pilot	Pilot 2	80-120	Owner + Founding Member 1
Release governance and rollback checklists	Pre-Pilot	Production	40-60	Technical Head
Staffing calibration and refresher training	Pilot 1	Production	40-60	Founding Member 1
Launch analytics and reporting setup	Pilot 2	Production	60-80	Technical Head
Compliance and regulatory documentation	Pre-Pilot	Production	80-120	Owner + Advisors

Total parallel operational effort: 620-880 hours (modeled at 750 hours expected)

17.2 Why This Matters for Timeline

The company is not only finishing software. It is converting a broad prototype into a disciplined healthcare operation. The parallel workstreams consume significant founder and founding-team time, which is why:

- Lower coding effort does not erase operating work
- Timeline compression has a floor determined by operational readiness
- The saved engineering time is partly consumed by QA, SOP writing, training, partner activation, and release governance

17.3 Combined Effort Summary

Workstream Category	Hours	Cost at Blended Rates
Web platform hardening	745	$14,900 (at $20/hr senior rate)
Android APK	480	$7,200 (at $15/hr blended)
QA, DevOps, security hardening	220	$4,400 (at $20/hr senior rate)
Parallel operational workstreams	750	$11,250 (at $15/hr blended)
Total	**2,195**	**$37,750**

Appendix: Key Differences from v1 Params

Parameter	v1 Value	v2 Value	Reason for Change
Team size	"Small team of 3" abstraction	3 core + 7 devs + 2 advisors = 12	Real team composition disclosed
Employment status	Implied full-time	All part-time (~4 hrs/day)	Actual working arrangement
Junior dev rate	Not specified	$10/hr	Founder-approved rate card
Senior dev rate	$24-36/hr range	$20/hr	Founder-approved rate card
Management rate	Blended	$20/hr	Founder-approved rate card
Everyone else	Blended at $30	$15/hr	Founder-approved rate card
Web effort	682-1,413 hrs (3-point estimate)	720-768 hrs (150-160% of Android)	Real team estimate
Android effort	440-640 hrs	480 hrs fixed	Contracted quote
MBBS doctor salary	INR 50,000-75,000	INR 60,000 (mid-range)	Verified against telemedicine benchmarks
Infrastructure	$30-50/mo	$65-1,500/mo by phase	Added phones, telecom, callback tooling, devices
Equity split	Not detailed	60/20/20	Founder-defined three-pool structure
Operational staff	Not explicitly modeled	Doctors + Admin + Tech Support ladder	Required for operating a healthcare company
Parallel workstreams	Not tracked	750 hours explicitly modeled	Company-building work beyond coding

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End of params.md — all downstream v2 documents reference this file for assumptions.